M.TECH. COMPUTER SCIENCE AND ENGINEERING – January - May, 2026

CS606 – Data Science and Artificial Intelligence Laboratory

#Session: 08 || Date: 23/03/2026

Time Series Analysis using Python and Tableau

1. Perform Time Series Analysis using for a dataset of your choice from the web, for the following models using the metrics given.
   Note: The dataset should be in a table format. You are not supposed to choose that dataset for which the code for the plots is available. You should be able to infer what a particular chart/plot reveals and explain it.
   a. AutoRegression
   b. Moving Average
   c. Single and Double Exponential Smoothing
   d. AutoRegressive Moving Average (ARMA)
   e. Box-Jenkins AutoRegressive Integrated Moving Average (ARIMA)
   f. Create the relevant graphs in Tableau for the visualization of Time Series Forecasting using the models mentioned in a - e.

Metrics: Akaike Information criterion (AIC) and Bayesian Information Criterion (BIC)

Learning Resources:

a. https://www.tableau.com/analytics/what-is-time-series-analysis

b. https://www.kaggle.com/code/prashant111/complete-guide-on-time-series-analysis-in-python

c. https://www.udacity.com/blog/2024/12/introduction-to-time-series-analysis-techniques-and-use-cases-in-data-science.html

d. https://www.ibm.com/think/topics/autoregressive-model

Moodle Submission Logistics:

- You need to have the following files in a zipped folder with the naming format:
  “TSA_<<roll_no>>”
  (a) Your code files along with graphs as the outcome with proper naming extension
  (b) Your dataset/s
  (c) any reference materials you have used.

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